Class activity: confidence intervals

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

Sparrows data

We are interested in whether there is a relationship between wing length (in mm) and weight (in g) for sparrows, using a sample of 116 sparrows. Our model is

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

Here is output for the fitted model in R:

```
            Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

1. Use the `qt` function in R to find the critical value for a 98% confidence interval:

```r
qt(..., df = ..., lower.tail = FALSE)
```

2. Calculate and interpret a 98% confidence interval for the average change in wing length associated with an increase of 1g in sparrow weight.

3. Reading the confidence interval, your friend claims that there is a 2% chance that the true slope β_1 is greater than 1.54 or less than 1.08. Is your friend correct?